

TECHNICAL SPECIFICATION

I.R.O.

13M/11M

WPB 160X160X30.44

WIDE FLANGE BEAM

WBSEDCL

1. SCOPE:

This specification covers design, manufacture, testing and supply of 13m/11M long WPB 160X160X30.44 Wide Flange Beam.

2. STANDARDS:

The Wide Flange Beam shall comply with the requirements of IS 12778:2004 and IS 2062:2011 with Grade Designation: E300 (Tensile strength: 440N/mm²) and Quality: "A" in respect of "dimensions and section properties" and steel specifications respectively.

3. DIMENSIONS AND SECTIONAL PROPERTIES:

Designation	Mass	Area	Depth	Width	Thickness		Root radius	Moment of Inertia		Radius of Gyration		Sectional modulus	
					Web	Flange							
	M	a	D	B	t	T	R	I _x	I _y	r _x	r _y	Z _x	Z _y
	Kg/m	Cm ²	mm	mm	mm	mm	mm	Cm ⁴	Cm ⁴	Cm	Cm	Cm ³	Cm ³
WPB 160X160X	30.44	38.8	152	160	6.0	9.0	15	1673	615.6	6.57	3.98	220.1	76.9

4. ROLLING AND CUTTING TOLERANCE:

4.1 Dimensional Tolerance

Designation	Depth	Width	Thickness	
			Web	Flange
	D	B	t	T
	mm	mm	mm	mm
WPB 160X160X30.44	+/- .3	+/- .3	+/- .0.7	+/- .1.5

Other dimensional tolerance will be in conformity to IS 12779:1989.

4.2 Mass Tolerance

The tolerance on mass per metre of individual section shall be +/- .4 percent of the specified mass i.e 30.44Kg/m. Mass tolerances for lot however shall be within +/- .3 percent of the specified values.

4.3 Cutting Tolerance

Cutting tolerance shall be within + 100, - 0 mm from the specified values.

5. EMBOSsing ON EACH JOIST:

Following Distinct non-erasable embossing to be made on each Wide Flange Beam:

- Name & Logo of the manufacturer.
- ISI mark.
- Designation of steel and Size of the Wide Flange Beam.

FOR

1. PARALLEL FLANGE STEEL SECTIONS:

Sl.no.	Particulars	WPB 160X160X30.44	Manufacturer's Data
1	Length of Joist in Metre.	13Mtr/11 Mtr	
2	Weight in Kg/m	30.44 kg/m	
3	Depth (D) of section in mm	152 mm	
4	Width(W) of section in mm	160 mm	
5	Thickness of flange(T) in mm	9.0 mm	
6	Thickness of web (t) in mm	6.0 mm	
7	Root radius (R) in mm	15 mm	
8	Moment of Inertia in cm ⁴ I _x I _y	1673 cm ⁴ 615.6 cm ⁴	
9	Radius of Gyration in cm r _x r _y	6.57cm 3.98 cm	
10	Sectional modulus in cm ³ Z _x Z _y	220.1 cm ³ 76.9 cm ³	

Sl.no.	Test	Requirement as per IS 2062:2011 with Grade Designation: E300 (Tensile Strength: 440N/mm ²) and Quality: "A"	Manufacturer's Data
A	Tensile Test:		
	i. Yield Stress (MPa)	Min. 300	
	ii. Tensile Strength(MPa)	Min. 440	
	iii. Percentage Elongation A , Min at Gauge Length, $L_0=5.65$	22	
B	Bend test: The test piece at room temperature shall withstand bending without cracking through 180° to an internal diameter not greater than:	Internal Bend Diameter: 2 X Thickness of test piece.	

Sl.no.	Test	Requirement as per IS 2062:2011 with Grade Designation: E300(Tensile strength: 440N/mm ²) and Quality: "A"	Manufacturer's Data
1	Carbon%	0.2	
2	Manganese%	1.5	
3	Sulphur%	0.045	
4	Phosphorous%	0.045	
5	Silicon%	0.45	
6	Carbon equivalent %(Max.)	0.44	
7	Mode of De-oxidation	Semi-killed or killed	

****This form is to be duly filled in by the Bidder and submitted along with the Tender**